

Retrieval-Augmented Generation (RAG)

What is RAG?

Retrieval-Augmented Generation is a pattern for question answering in which a system first retrieves relevant information from a knowledge source and then gives that information to a language model as context for generating an answer.

RAG Pipeline

A basic pipeline is: documents -> text extraction -> chunking -> embeddings -> vector storage -> similarity search -> retrieved context -> language model -> answer. Each stage affects the quality of the final response.

PDF Ingestion

For PDF-based RAG, the ingestion stage reads a PDF and extracts its text along with useful metadata such as file name and page number. Clean, searchable PDFs are generally easier to ingest than scanned PDFs, which may require OCR.

Chunking

Large documents are divided into smaller chunks before embedding. Chunk size and overlap should preserve enough context for retrieval while keeping each chunk focused. Poor chunking can cause important information to be split or unrelated information to be merged.

Embeddings and Vector Search

An embedding converts text into a numerical vector that represents semantic information. During retrieval, a user query is embedded and compared with stored vectors using a similarity measure. The highest-scoring chunks are typically supplied to the language model.

Example Query

Question: "What is the purpose of chunking in a RAG pipeline?" Relevant evidence may explain that chunking creates smaller searchable units, improves focused retrieval, and helps fit useful context into the model input.

Common Problems

RAG systems can fail because extraction is incomplete, chunks are poorly designed, retrieval returns irrelevant passages, metadata is lost, or the model answers beyond the retrieved evidence. Evaluation should test retrieval quality as well as final answer quality.